1a.import pandas as pd

```
data = pd.read_csv("https://raw.githubusercontent.com/mwaskom/seaborn-data/master/iris.csv")
print(data.head())
print(data.shape)
print(data.columns.tolist())
print(data.dtypes)
data.info()
print(data.describe())
print(data.isnull().sum())
```

1b. import pandas as pd

```
data = pd.read_csv("https://raw.githubusercontent.com/mwaskom/seaborn-data/master/iris.csv")
data = data.fillna(0).drop_duplicates()
data.columns = ["SepalLength", "SepalWidth", "PetalLength", "PetalWidth", "Species"]
print(data.head())
```

2.import pandas as pd

```
import matplotlib.pyplot as plt
df = pd.read_csv('1. Weather_Data.csv').dropna(axis=1)
df.rename(columns={'Weather':'Weather_Condition'}, inplace=True)
print(df.shape, df.isnull().sum())
print(df['Weather_Condition'].value_counts())
print(df.describe())
df.plot(x='Date/Time', y='Temp_C')
plt.show()
df['Weather_Condition'].value_counts().loc[lambda x:x>20].plot.pie(autopct='%1.1f%%')
plt.show()
```

3.import os, shutil

```
os.makedirs("HDFS/input", exist_ok=True)
os.makedirs("HDFS/output", exist_ok=True)
```

```
os.makedirs("HDFS/backup", exist_ok=True)
with open("HDFS/input/data.txt", "w") as f:
    f.write("Welcome to HDFS")
shutil.copy("HDFS/input/data.txt", "HDFS/backup/")
shutil.move("HDFS/input/data.txt", "HDFS/output/")
os.rename("HDFS/output/data.txt", "HDFS/output/final.txt")
for r, d, f in os.walk("HDFS"):
    print(r, d, f)
shutil.rmtree("HDFS")
```

```
4import pandas as pd
df = pd.read_csv("https://raw.githubusercontent.com/mwaskom/seaborn-
data/master/iris.csv").drop_duplicates()
print(df.head(), df.info(), df.describe(), df.isnull().sum())
print(df[df.species == 'setosa'].head())
print(df.sort_values('sepal_length').head())
print(df.groupby('species').sepal_length.agg(['count', 'mean']))
df['SepalArea'] = df.sepal_length * df.sepal_width
df.to_csv("Processed_Iris.csv", index=False)
```

```
5import json
f = "Students.json"
while True:
    try:
        d = json.load(open(f))
    except:
        d = []
    c = input("\n1.Add 2.Read 3.Update 4.Delete 5.Exit: ")
    if c == "1":
        d.append({"ID": input("ID: "), "Name": input("Name: "),
                  "Age": input("Age: "), "Course": input("Course: ")})
```

```
 elif c == "2":
        print(d)
    elif c in ["3", "4"]:
        sid = input("Enter ID: ")
        s = next((x for x in d if x["ID"] == sid), None)
        if s:
            if c == "3":
                s["Name"] = input("New name: ")
            else:
                d.remove(s)
        else:
            print("Not found")
    elif c == "5":
        break
    else:
        print("Invalid choice")
        continue
    if c != "2":
        json.dump(d, open(f, "w"), indent=4)
```

6.

```
import json
students = [
    {"ID":101,"Name":"Shashank","Age":22,"Course":"CSE"},
    {"ID":102,"Name":"Vinay","Age":21,"Course":"ECE"},
    {"ID":103,"Name":"Ram","Age":23,"Course":"CSE"}
]
json.dump(students, open("students.json","w"), indent=4)
data = json.load(open("students.json"))
print(data)
name = input("Enter name: ")
print([s for s in data if s["Name"].lower() == name.lower()])
```

```
course = input("Enter course: ")
print([s for s in data if s["Course"].lower() == course.lower()])
age = int(input("Minimum age: "))
print([s for s in data if s["Age"] >= age])
```

7.
```
n = int(input("Number of lines: "))
words = []
for _ in range(n):
    words += input().lower().split()
print("Map:", [(w, 1) for w in words])
count = {w: words.count(w) for w in set(words)}
print("Reduce (Word Count):", count)
```

8.
```
import pandas as pd
users = pd.DataFrame({
    'action':['login','logout','login'],
    'timestamp':['10:00','10:15','10:30']
})
print("Hive:\n", users.groupby('action').size())
print(users)
logs = pd.DataFrame({'type':['info','error','error']})
print("Pig:\n", logs[logs.type=='error'].groupby('type').size())
print(logs)
```

9.

first run this

```
%pip install pyspark
```

code:

```
from pyspark.sql import SparkSession
from pyspark.sql.functions import *
spark = SparkSession.builder.appName("Students").getOrCreate()
data = [(101,"Rahul","CSE",85), (102,"Priya","ISE",90),
```

```
    (103,"Arun","CSE",78), (104,"Sneha","ECE",88)]
df = spark.createDataFrame(data, ["ID","Name","Department","Marks"])
df.show()
df.filter(df.Marks > 80).show()
df.groupBy("Department").avg("Marks").show()
df.agg(max("Marks"), min("Marks")).show()
spark.stop()
```

10.

```
import pandas as pd
import matplotlib.pyplot as plt
from sklearn.linear_model import LinearRegression
df = pd.read_csv("https://raw.githubusercontent.com/mwaskom/seaborn-data/master/iris.csv")
X, y = df[["petal_length"]], df["sepal_length"]
m = LinearRegression().fit(X, y)
p = m.predict(X)
print(p[:5])
plt.scatter(X, y)
plt.plot(X, p, "r")
plt.show()
```